PIDG: MLP Capacity and Missing Frames

Fixed 50-view official test | 2026-10-21 12:34 BST

> **One evaluation set:** every PSNR, SSIM, LPIPS and MS-SSIM result uses the same 50 official test images. Metrics are computed per image and averaged.

Both studies use the original PIDG compatibility baseline: Rubber Duck, 50k iterations, 1600×900 renders and training seed0. The capacity study changes the deformation MLP; the removal study keeps original W256_D0 and removes training images.

> **Capacity:** W512_D0 has the highest PSNR, 28.577 dB, a 1.320 dB gain over baseline. Larger capacity is not monotonically better; original baseline has the lowest LPIPS.

> **Training-frame removal:** 10% yields 28.253 dB; 90% falls to 23.063 dB, 4.194 dB below the complete-training-data baseline. Intermediate ratios are not strictly monotonic.

One scene, one training seed and one removal seed; historical baseline reused. These are descriptive results without a statistical significance claim.

Guide: architecture 02; protocol 03; MLP results 04-05; removal results 06-08; snapshots and sources 09; absolute render / GT paths 10-11.

What Changes in the Deformation MLP?

Width controls branch size; depth counts extra shared hidden layers after fusion.

Module	Structure
Spatial branch	XYZ hash: 32 → Linear(32, w) → 2×Sigmoid−1
Temporal branch	XYT / YZT / XZT hash: 192 → Linear(192, w) → ReLU
Fusion	Elementwise product of both branches; w features
Extra shared layers	D0: none; D2: two Linear(w, w) → ReLU blocks
Three output heads	w→7: positional warp (quaternion (4) + translation (3)); w→4: rotation offset; w→3: scale offset

Model	w / d	MLP parameters	Relative size
W256_D0	256 / 0	61,454	1.00×
W512_D0	512 / 0	122,894	2.00×
W256_D2	256 / 2	193,038	3.14×
W512_D2	512 / 2	648,206	10.55×

D0 still has a network: both input branches and all three linear output heads remain, with no extra hidden layer after fusion. W512_D0 widens 256 features to 512. Counts exclude hash grids.

Hash grids, Gaussians, renderer and velocity / stress network are unchanged. The original velocity / stress MLP has a shared 43→64→64 trunk with 3D velocity and 6-component stress heads. New F / NH / material / structured-force branches are disabled.

Fixed Test Set, Different Training Subsets

Removal drops whole training images, not pixels; removal seed0, nested subsets.

Removed	Training kept	Training removed	Test images
0%	150	0	50
10%	135	15	50
30%	105	45	50
50%	75	75	50
70%	45	105	50
90%	15	135	50

The original split has 150 training frames and 50 fixed official test frames. Only the training manifest changes. Images, times, poses and canonical points remain. Higher removal includes earlier removed frames; original order is preserved.

> **Test-set usage:** retained training images supply the loss and gradients. Official test images do not supply gradient updates, but are evaluated during training to select ours_best by test PSNR. The test set therefore also serves model selection.

Only best-snapshot PNG metrics on the fixed official test set are compared. New runs train to 50k; best may come from an earlier iteration. Higher PSNR, SSIM and MS-SSIM are better; lower LPIPS is better.

A fixed 50k budget increases image reuse at high removal. The denominator of native PIDG momentum queries, fid/train_count, also changes with training-set size; original behavior is preserved. PSNR averages per-image dB values.

Wider Helps; Capacity Gains Are Not Monotonic

The same 50 official test images; best-snapshot PNG renders.

Model	Parameters	PSNR	ΔPSNR	SSIM	LPIPS	MS-SSIM
W256_D0	61,454	27.257	+0.000	0.976551	0.029624	0.970483
W512_D0	122,894	28.577	+1.320	0.977049	0.034613	0.970466
W256_D2	193,038	28.110	+0.853	0.976506	0.031238	0.970350
W512_D2	648,206	27.615	+0.359	0.976784	0.033296	0.971072

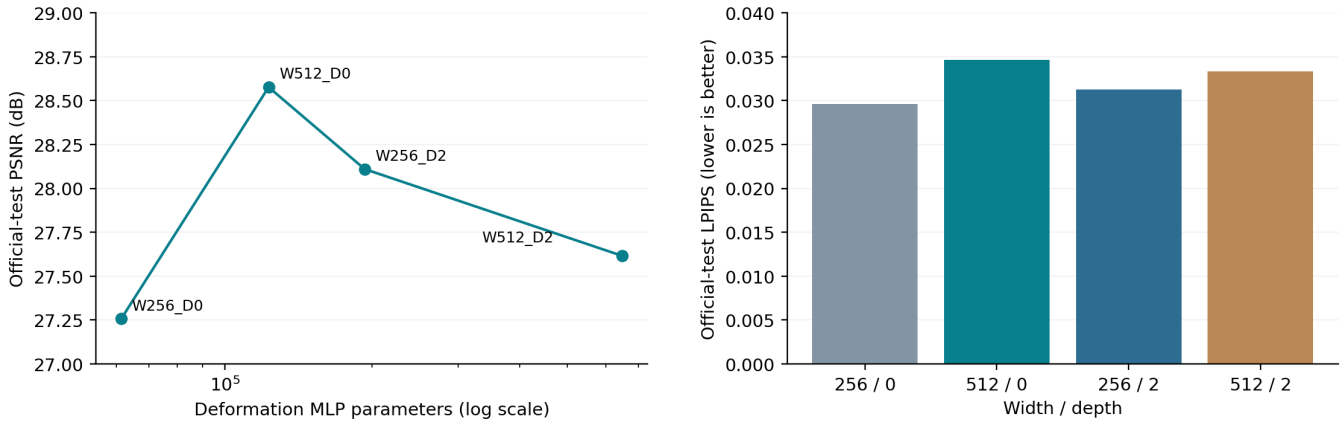


Figure 1. Capacity versus PSNR / LPIPS. Connecting lines guide reading; the parameter axis is logarithmic.

W512_D0 gains 1.320 dB over baseline. At width 512, adding two shared hidden layers reduces PSNR by 0.961 dB. The largest model does not achieve the highest PSNR.

W256_D0 has the lowest LPIPS; W512_D2 has the highest MS-SSIM. Rankings depend on the metric. Historical baseline lacks complete source hashes; one seed cannot establish stable causal gains.

MLP Capacity: GT and Four Models

Official test index 25 | source train/0101 | t=0.510

The same GT-derived crop and proportional resizing are used for display only. No color enhancement; metrics use complete 1600×900 PNGs.

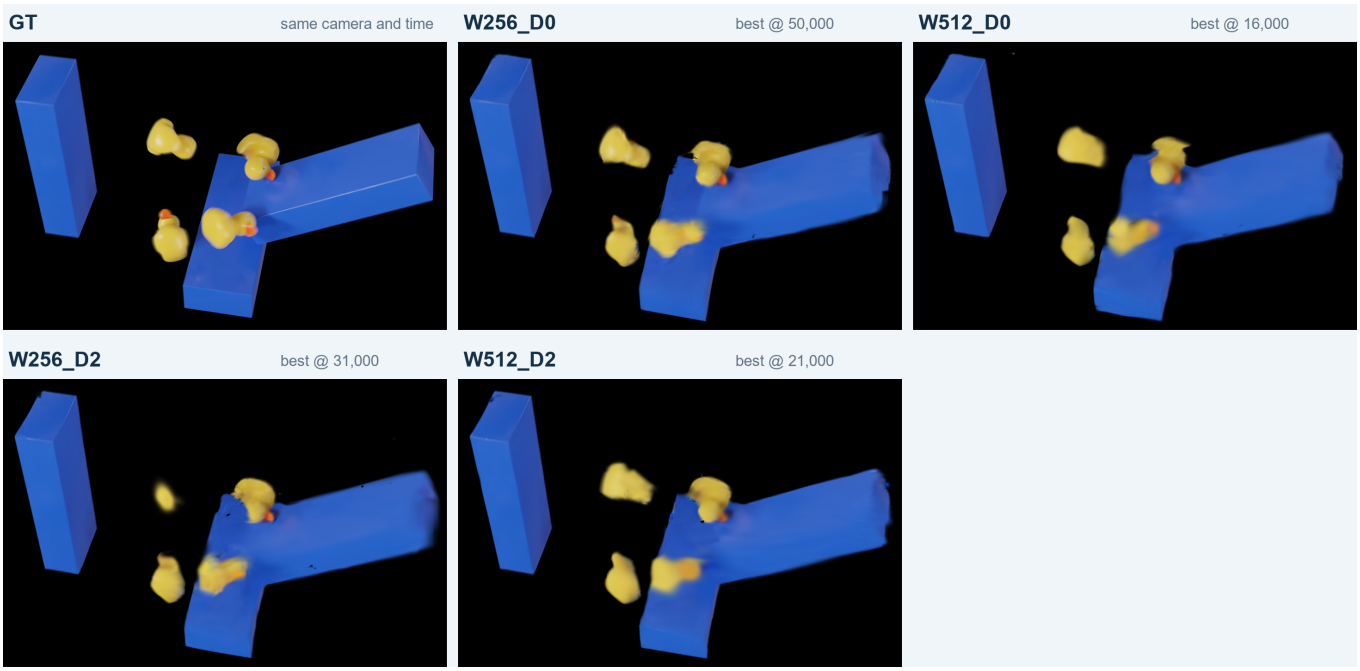


Figure 2. GT and four deformation MLPs, labelled by best iteration. The final empty cell is a layout choice.

Compare contours, occlusions and support edges. One view cannot replace the 50-image mean. Absolute MLP render and GT directories are on page 10.

Severe Removal Reduces Reconstruction Quality

Every model uses W256_D0; evaluation always uses the same 50 official test images.

Removed	Train kept	PSNR	ΔPSNR	SSIM	LPIPS	MS-SSIM
0%	150	27.257	+0.000	0.976551	0.029624	0.970483
10%	135	28.253	+0.996	0.976569	0.029416	0.970624
30%	105	26.686	-0.571	0.975140	0.031599	0.968252
50%	75	27.043	-0.214	0.973506	0.034204	0.964774
70%	45	26.930	-0.326	0.973107	0.035995	0.964997
90%	15	23.063	-4.194	0.953808	0.060427	0.936079

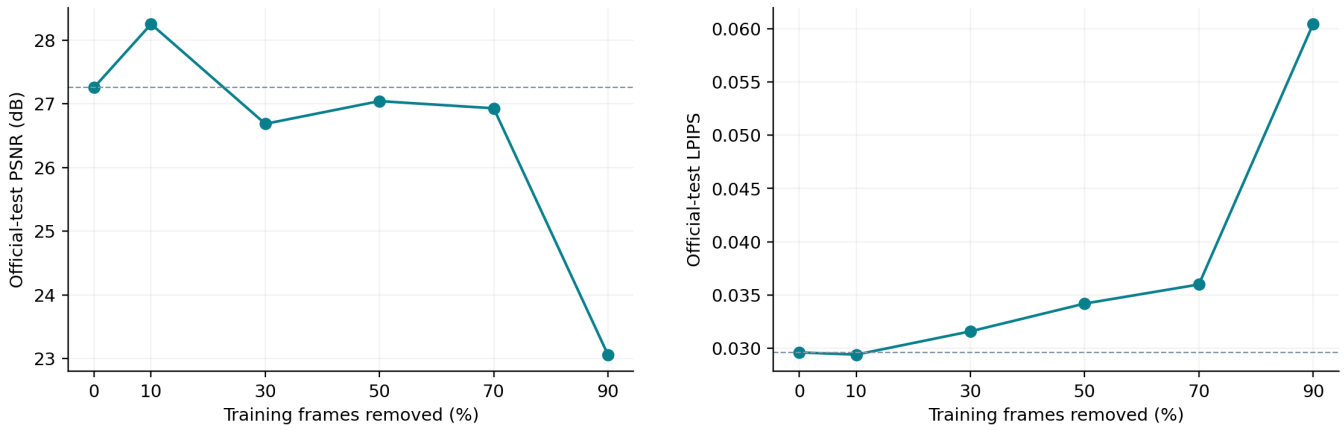


Figure 3. Removal curves on the same official test set; dashed lines mark the complete-training-data baseline.

10% removal exceeds historical baseline by 0.996 dB; intermediate ratios are not strictly monotonic. At 90%, PSNR is 4.194 dB below baseline and LPIPS rises from 0.029624 to 0.060427.

The single-run 10% gain needs matched-environment repeats over training and removal seeds. It does not establish that deleting training images is generally beneficial. Test membership is identical across all six ratios.

Official-Test Images: 0%, 10%, 30%

Official test index 25 | source train/0101 | t=0.510

GT and models use the same camera, time and crop. Cropping is display-only; metrics use complete images.

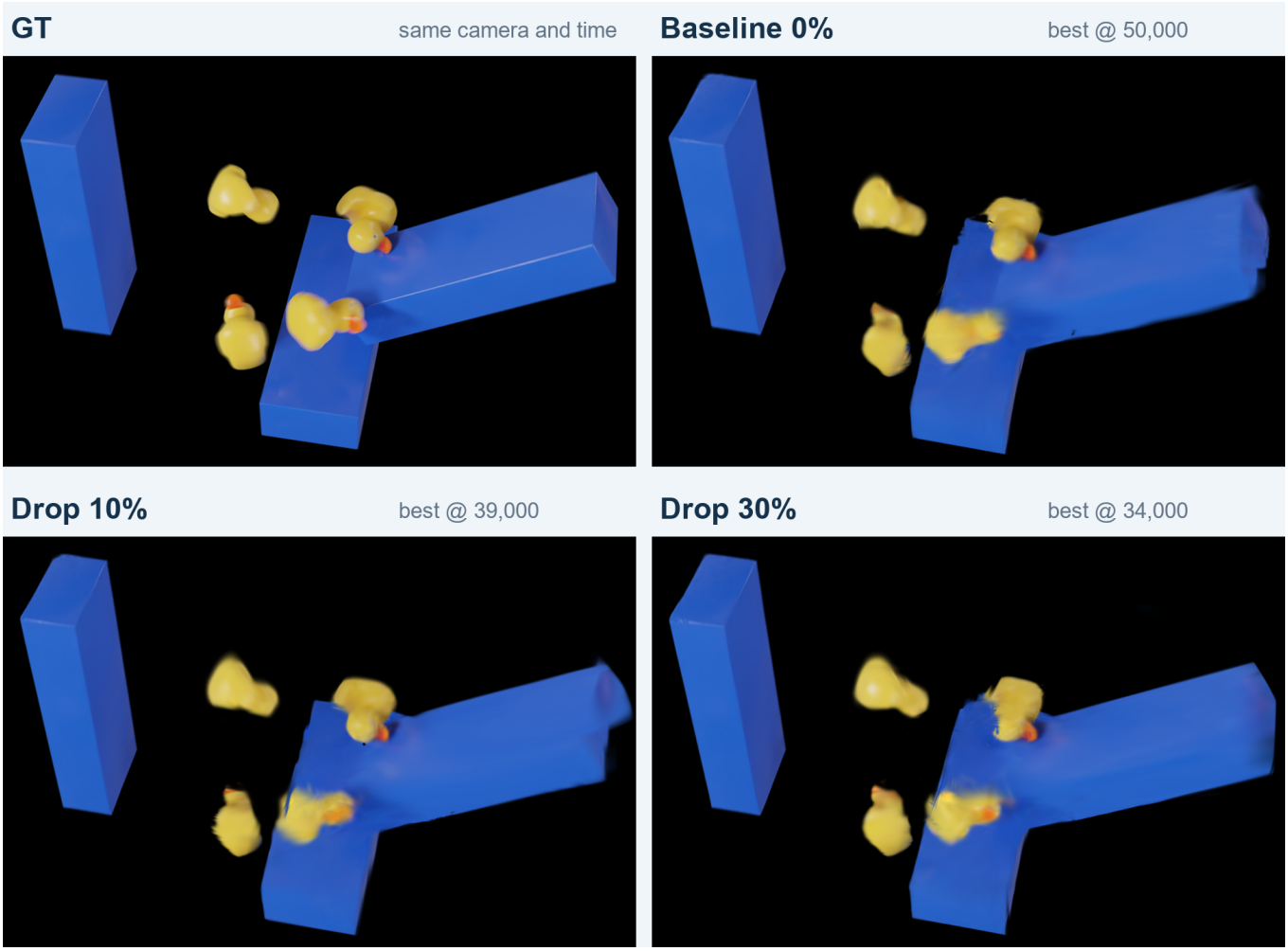


Figure 4. A fixed official test view, labelled by each model's best iteration.

Compare shape, occlusion, support edges and detail. Numerical conclusions use the 50-image means on page 6. Absolute removal-model paths are on page 11; the 0% control is on page 10.

Official-Test Images: 50%, 70%, 90%

Official test index 25 | source train/0101 | t=0.510

GT and models use the same camera, time and crop. Cropping is display-only; metrics use complete images.

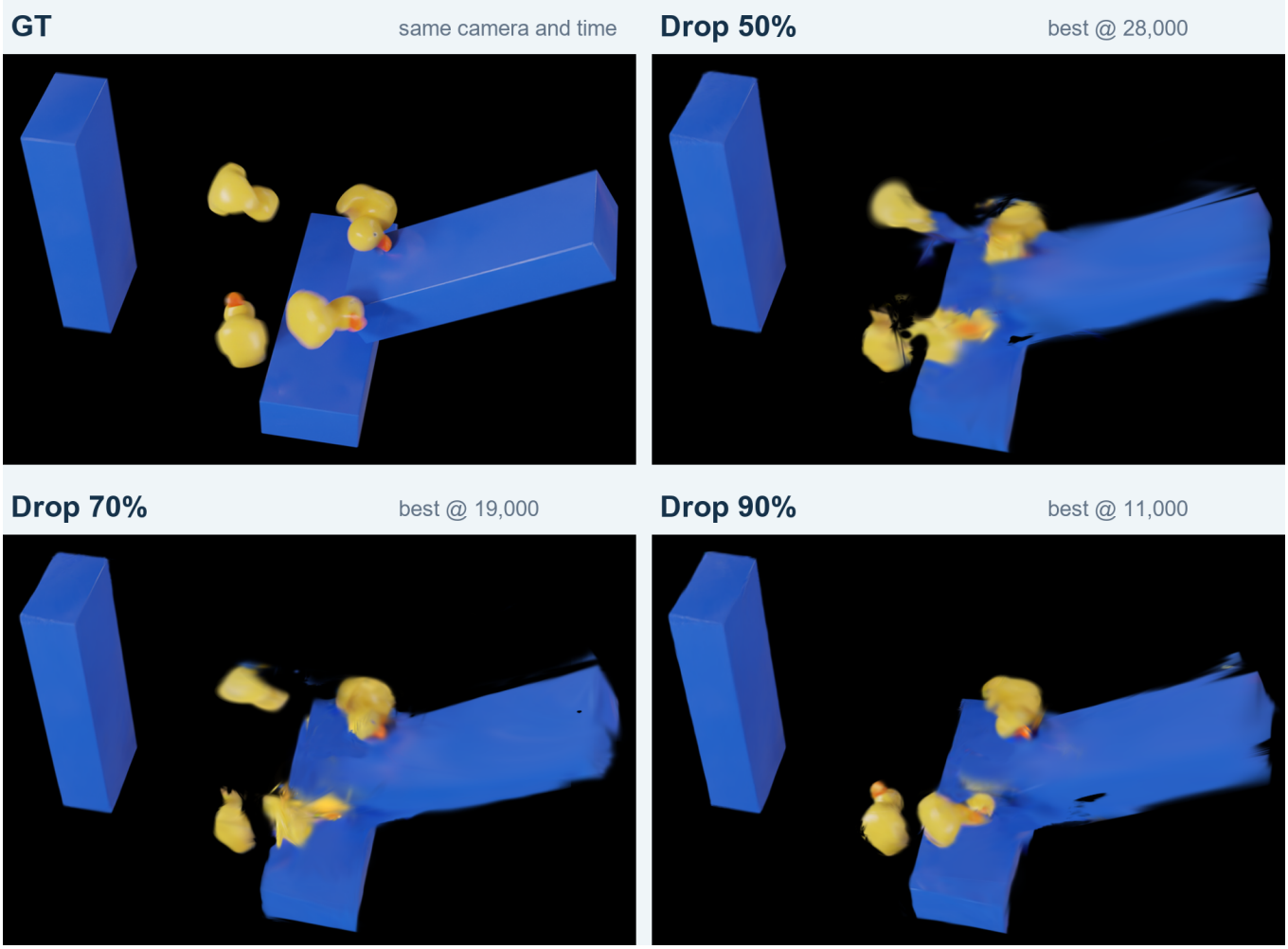


Figure 5. A fixed official test view, labelled by each model's best iteration.

Compare shape, occlusion, support edges and detail. Numerical conclusions use the 50-image means on page 6. Absolute removal-model paths are on page 11; the 0% control is on page 10.

Snapshots, Limits and Sources

50k is the completed training budget; the table lists the best snapshots actually rendered.

Model	Best iteration
W256_D0	50,000
W512_D0	16,000
W256_D2	31,000
W512_D2	21,000
Removal 10%	39,000
Removal 30%	34,000
Removal 50%	28,000
Removal 70%	19,000
Removal 90%	11,000

Limits: 90% removal leaves 15 training images covering roughly t=0.067 to 0.919. Official testing includes times outside that interval, so temporal extrapolation contributes. Canonical points remain shared.

Comparability: all results use the same 50 official test images. Historical baseline reused; training seed0 and removal seed0. Official-test PSNR selects best, so the reported test set is not independent of model selection. Statistical significance is not established.

Local sources for verification

experiments/mlp_capacity/{results.json,plan.json,validation.json}

experiments/random_frame_mask/{results.json,plan.json,preflight.json}

third_party/PIDG/scene_PIDG/network.py; train_pidg.py; render.py; metrics.py

data/PIDG/rubber_duck/transforms_test.json; run files: reproduction.json, results.json, per_view.json

Companion sources.json records official-test metrics, best iterations, image paths and source hashes. All 9 unique render directories contain 50 PNGs and matching GT.

MLP Study: Absolute Render and GT Paths

All directories contain the fixed 50 official test views. Line breaks are layout only; no path characters are omitted.

W256_D0 | best 50,000

Render:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_official\rubber_duck\test\ours_best\renders
```

GT:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_official\rubber_duck\test\ours_best\gt
```

W512_D0 | best 16,000

Render:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_mlp_capacity\long_W512_D0\rubber_duck\test\ours_best\renders
```

GT:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_mlp_capacity\long_W512_D0\rubber_duck\test\ours_best\gt
```

W256_D2 | best 31,000

Render:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_mlp_capacity\long_W256_D2\rubber_duck\test\ours_best\renders
```

GT:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_mlp_capacity\long_W256_D2\rubber_duck\test\ours_best\gt
```

W512_D2 | best 21,000

Render:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_mlp_capacity\long_W512_D2\rubber_duck\test\ours_best\renders
```

GT:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_mlp_capacity\long_W512_D2\rubber_duck\test\ours_best\gt
```

Each directory contains 00000.png through 00049.png. Render and GT filenames match at 1600×900 resolution.

Removal Study: Absolute Render and GT Paths

All directories contain the fixed 50 official test views. Line breaks are layout only; no path characters are omitted.

Drop 10% | best 39,000

Render:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_random_frame_mask\long_drop10_seed0\rubber_duck\test\ours_best\renders
```

GT:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_random_frame_mask\long_drop10_seed0\rubber_duck\test\ours_best\gt
```

Drop 30% | best 34,000

Render:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_random_frame_mask\long_drop30_seed0\rubber_duck\test\ours_best\renders
```

GT:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_random_frame_mask\long_drop30_seed0\rubber_duck\test\ours_best\gt
```

Drop 50% | best 28,000

Render:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_random_frame_mask\long_drop50_seed0\rubber_duck\test\ours_best\renders
```

GT:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_random_frame_mask\long_drop50_seed0\rubber_duck\test\ours_best\gt
```

Drop 70% | best 19,000

Render:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_random_frame_mask\long_drop70_seed0\rubber_duck\test\ours_best\renders
```

GT:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_random_frame_mask\long_drop70_seed0\rubber_duck\test\ours_best\gt
```

Drop 90% | best 11,000

Render:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_random_frame_mask\long_drop90_seed0\rubber_duck\test\ours_best\renders
```

GT:

```
F:\A_Edinburgh\PhysGauss\runs\pidg_random_frame_mask\long_drop90_seed0\rubber_duck\test\ours_best\gt
```

Each directory contains 00000.png through 00049.png. Render and GT filenames match at 1600×900 resolution.

0% removal reuses W256_D0; its official-test paths are the first entry on page 10.